CUSTOMER PERCEPTION OF AI-POWERED PERSONALIZED RECOMMENDATIONS ON SOCIAL MEDIA PLATFORMS

PROJECT REPORT

by

ANANYA GOPALAKRISHNAN

Contents

1 INTRODUCTION **6**

1.1 Introduction 7
1.2 Statement of the Problem 8
1.3 Objectives of the Study 8
1.4 Significance of the Study 8
1.5 Scope of the Study 9
1.6 Limitations of the Study 9
1.7 Structure of the Report 9

2 RESEARCH METHODOLOGY **11**

2.1 Introduction 12
2.2 Research Design 12
2.3 Nature of Study 12
2.4 Source of Data 12
2.5 Sampling Design 12
2.5.1 Population of the Study 12
2.5.2 Sampling Technique 13
2.5.3 Sample Size Determination Using Cochran's Formula 13
2.5.4 Proportionate Allocation 14
2.5.5 Sampling Unit 14
2.6 Data Collection Instrument 14
2.7 Measurement Scale 14
2.8 Variables Used in the Study 15
2.9 Tools Used for Analysis 16
2.9.1 Statistical Tools 16
2.9.2 Software Tools 17
2.9.3 Analytical Tools Used Based on Objectives 17
2.10 Limitations of the Study 17
2.11 Chapter Summary 18

3 THEORETICAL FRAMEWORK **19**
3.1 Introduction 20
3.2 Artificial Intelligence-Based Personalized Recommendations 20
3.3 Technology Acceptance Model (TAM) 22
3.4 Construct Variables of the Study 22
3.4.1 Accuracy 22
3.4.2 Privacy 23
3.4.3 Usefulness 23
3.4.4 Ease of Use 24
3.4.5 Behavioural Intention 24
3.4.6 Customer Perception 25
3.5 Proposed Research Model 25
3.6 Hypotheses Development 25
3.7 Conceptual Framework 26
3.8 Chapter Summary 26

4 DATA ANALYSIS AND INTERPRETATION **27**
4.1 Introduction 28
4.2 Data Collection Methodology 28
4.2.1 Survey Instrument Design 28
4.2.2 Population of the Study 29
4.2.3 Sampling Technique, Sample Size Determination & Proportionate Allocation 29
4.2.4 Data Collection & Data Preparation 31
4.3 Demographic Profile of Respondents 32
4.3.1 Age Group Distribution 32
4.3.2 Gender Distribution 33
4.3.3 Education Distribution 34
4.4 Influence of Accuracy, Privacy and Usefulness on Customer Perception: Regression Analysis 35
4.4.1 Introduction 35
4.4.2 Hypothesis Formulation 35
4.4.3 Statistical Output 36
4.4.4 Fitted Regression Model 36
4.4.5 Interpretation 36
4.4.6 Correlation Analysis HeatMap 37
4.4.7 Conclusion 39
4.5 Influence of Demographic Factors in Customer Perception 39

4.5.1 Introduction 39
4.5.2 Hypothesis Formulation 40
4.5.3 Statistical Output 40
4.5.4 Interpretation 41
4.5.5 Visual Representation 41
4.5.6 Conclusion 43
4.6 Influence of Ease of Use on Customer Perception 43
4.6.1 Introduction 43
4.6.2 Hypothesis Formulation 44
4.6.3 General Regression Model 44
4.6.4 Summary Statistics for Ease of Use 44
4.6.5 Statistical Output 44
4.6.6 Fitted Regression Model 45
4.6.7 Interpretation 45
4.6.8 Visual Representation 46
4.6.9 Conclusion 46
4.7 Impact of Perceived Factors on Behavioural Intention 47
4.7.1 Introduction 47
4.7.2 Hypothesis Formulation 47
4.7.3 Statistical Output 47
4.7.4 Fitted Regression Model 48
4.7.5 Interpretation 49
4.7.6 Visual Representation 49
4.7.7 Conclusion 50
4.8 Chapter Summary 50

5 FINDINGS, CONCLUSIONS AND SUGGESTIONS 52
5.1 Introduction 53
5.2 Major Findings 53
5.3 Overall Findings 54
5.4 Conclusion 54
5.5 Suggestions 54
5.6 Scope for Future Research 55
5.7 Limitations of the Study 55
5.8 Chapter Summary 55

Appendix 57

List of Tables

4.1 Age Group Distribution with Sample Size 31
4.2 Age Distribution of Respondents 32
4.3 Gender Distribution of Respondents 33
4.4 Educational Qualification of Respondents 34
4.5 Regression Analysis Results 36
4.6 Model Summary Statistics 36
4.7 ANOVA Table for Age Group 40
4.8 t-test Results for Mean Perception by Gender 40
4.9 ANOVA Table for Education 41
4.10 Descriptive Statistics 44
4.11 Model Summary 44
4.12 ANOVA Summary 45
4.13 Regression Results 45
4.14 Model Summary 47
4.15 ANOVA Summary for Regression Model 48
4.16 Regression Coefficients 48

List of Figures

4.1 Age Distribution 32
4.2 Gender Distribution 33
4.3 Gender Distribution 34
4.4 Correlation HeatMap 37
4.5 BoxPlot 41
4.6 BoxPlot 42
4.7 BoxPlot 42
4.8 Scatter Plot 46
4.9 Scatter Plot 49

Chapter 1

INTRODUCTION

1.1 Introduction

Artificial Intelligence (AI) has become an integral part of modern digital platforms, transforming how users interact with online systems and services. One of the most prominent applications of AI is personalized recommendation systems, which analyze user data to provide tailored suggestions based on individual preferences, behavior, and past interactions. These systems are widely used across various domains such as e-commerce platforms, streaming services, social media, online learning platforms, and digital marketing applications.

AI-based personalized recommendations enhance user experience by delivering relevant content, products, or services. These systems help users make decisions quickly by filtering large volumes of information and presenting customized suggestions. For businesses, personalized recommendations improve customer engagement, increase conversion rates, and strengthen customer relationships. As a result, organizations increasingly rely on AI-driven recommendation systems to gain competitive advantage and improve customer satisfaction.

Despite the benefits, the adoption of AI-based personalized recommendation systems also raises certain concerns among users. Issues such as privacy risks, data security, transparency, and trust influence how users perceive AI-generated recommendations. Users may feel uncomfortable when platforms collect personal data to generate recommendations. Additionally, the ease of using such systems and users' behavioral intentions also play an important role in determining their acceptance of AI-based technologies.

Understanding user perception toward AI-based personalized recommendations is essential for organizations to improve system design and ensure user acceptance. Various factors such as perceived ease of use, privacy concerns, and behavioral intention influence users' perception and adoption of AI recommendation systems. Studying these factors helps organizations identify areas of improvement and develop more user-friendly and trustworthy systems.

This study aims to analyze user perception toward AI-based personalized recommendations and examine the factors influencing acceptance of these systems. The research focuses on identifying key variables affecting user perception and understanding how these variables impact users' attitudes and behavioral intentions toward AI-based recommendation systems.

The findings of this study will provide valuable insights for businesses, developers, and researchers to enhance AI-based recommendation systems and improve overall user experience.

1.2 Statement of the Problem

AI-based personalized recommendation systems are increasingly used to improve user experience. However, despite their advantages, users may have concerns regarding privacy, trust, and accuracy of recommendations. Some users may find AI recommendations helpful, while others may be hesitant due to privacy risks or lack of transparency.

Therefore, it is important to examine how users perceive AI-based personalized recommendations and identify the factors influencing their acceptance and usage. This study attempts to analyze user perception and determine key factors affecting adoption of AI-based recommendation systems.

1.3 Objectives of the Study

The main objectives of this study are:

1. To examine the influence of perceived accuracy, data privacy, and usefulness on customer perception of AI-based personalized recommendations.

2. To explore generational and demographic differences in customer perception of AI-based personalized recommendations.

3. To examine the influence of perceived ease of use on customer perception of AI-based personalized recommendations.

4. To analyze the impact of perceived factors (accuracy, privacy, usefulness, and perception) on customers' behavioural intention toward AI-based personalized recommendations.

1.4 Significance of the Study

This study is significant as it provides insights into user perception of AI-based personalized recommendations. The findings help organizations understand user expectations and improve AI recommendation systems. The study also contributes to academic research by analyzing factors influencing user acceptance of AI technologies.

The results of this study will benefit businesses, researchers, and developers by providing valuable insights into user preferences and concerns regarding AI-based personalized systems.

1.5 Scope of the Study

The study focuses on analyzing user perception towards AI-based personalized recommendations. The research examines factors such as privacy concerns, behavioral intention, and ease of use. The study is based on primary data collected through questionnaire responses from users familiar with AI-based recommendation systems.

1.6 Limitations of the Study

The study has the following limitations:

- The study is based on a limited sample size
- The responses are based on user perceptions which may vary
- The study focuses only on selected factors influencing user perception
- Time constraints limited the scope of data collection

1.7 Structure of the Report

This report is organized into eight chapters to systematically present the research work and findings.

- **Chapter 1: Introduction**
  This chapter provides an overview of the study, background of AI-based personalized recommendations, research objectives, significance of the study, and structure of the report.
- **Chapter 2: Literature Review**
  This chapter reviews previous studies related to AI-based personalized recommendations, customer perception, privacy concerns, ease of use, and behavioural intention. It helps identify research gaps and provides a foundation for the current study.
- **Chapter 3: Research Methodology**
  This chapter explains the research design, data collection methods, sampling technique, questionnaire design, variables used in the study, and statistical tools applied for analysis.
- **Chapter 4: Theoretical Framework**
  This chapter presents the theoretical foundation of the study based on the Technology Acceptance Model (TAM). It explains the relationship between perceived accuracy,

privacy, usefulness, ease of use, customer perception, and behavioural intention. The conceptual model and hypotheses are also discussed in this chapter.

- **Chapter 5: Data Analysis and Interpretation**
  This chapter presents the statistical analysis of the collected data using descriptive statistics, correlation analysis, regression analysis, ANOVA, and t-tests. The results are interpreted in relation to the research objectives.

- **Chapter 6: Findings, Conclusion, and Suggestions**
  This chapter summarizes the major findings of the study, provides conclusions based on the analysis, discusses practical implications, and offers suggestions for future research.

- **References**
  Lists all the academic sources, research papers, books, and websites referred to during the study.

- **Appendix**
  Includes supporting materials such as questionnaire, additional tables, charts, and other relevant documents used in the study.

Chapter 2

RESEARCH METHODOLOGY

2.1 Introduction

This chapter explains the research methodology adopted for the study. It describes the research design, population of the study, sampling technique, sample size determination, data collection method, and statistical tools used for analysis. The methodology provides a systematic approach to examine customer perception and behavioural intention toward AI-based personalized recommendations.
The study is entirely based on primary data collected from respondents through a structured questionnaire.

2.2 Research Design

The present study adopts a descriptive research design. Descriptive research design is used to describe customer perception and behavioural intention toward AI-based personalized recommendations. This design helps in understanding opinions, attitudes, and behaviour of respondents regarding AI-based recommendation systems.

2.3 Nature of Study

The present study is quantitative in nature. Quantitative research involves collecting numerical data and analysing it using statistical tools. A structured questionnaire was used to collect responses from respondents.

2.4 Source of Data

The study is based entirely on primary data. Primary data was collected directly from respondents through a structured questionnaire distributed online.

2.5 Sampling Design

2.5.1 Population of the Study

This study focuses on users of AI-based personalized recommendations in Malappuram district. Therefore, the population of the study comprises residents of Malappuram district. According to Census 2011, the total population of Malappuram district is 4,112,920.
The respondents include individuals who interact with AI-based personalized recommendations through digital platforms such as e-commerce websites, streaming platforms, and

social media.

2.5.2 Sampling Technique

Stratified random sampling method with proportionate allocation was adopted for the study. The population was divided into different age groups to ensure proper representation of respondents.
The age groups considered for the study are:

- 18–25 years
- 26–35 years
- 36 years and above

Stratified sampling ensures representation of respondents across different age groups and improves the reliability of the study.

2.5.3 Sample Size Determination Using Cochran's Formula

Cochran's Formula:

$$n_0 = \frac{Z^2 \times p \times q}{e^2}$$

Where:

- n_0 = Sample size
- Z = 1.96
- p = 0.5
- q = 0.5
- e = 0.05

$$n_0 = \frac{(1.96)^2 \times 0.5 \times 0.5}{(0.05)^2} = \frac{3.8416 \times 0.25}{0.0025} = \frac{0.9604}{0.0025} = 384.16$$

Thus, the ideal sample size is approximately 384 respondents.

However, due to time and resource constraints, the sample size was reduced to 250 respondents. This corresponds to an approximate margin of error of around 6%, which is acceptable for social science research.

2.5.4 Proportionate Allocation

The final sample size of 250 respondents was proportionately distributed:

- 18–25 years: $0.45 \times 250 = 113$
- 26–35 years: $0.30 \times 250 = 75$
- 36+ years: $0.25 \times 250 = 62$

2.5.5 Sampling Unit

The sampling unit consists of individuals residing in Malappuram district who use AI-based personalized recommendation systems.

2.6 Data Collection Instrument

A structured questionnaire was used. It included:

- Demographic variables
- Perceived Accuracy
- Privacy Concerns
- Usefulness
- Ease of Use
- Customer Perception
- Behavioural Intention

2.7 Measurement Scale

- Strongly Disagree
- Disagree
- Neutral
- Agree
- Strongly Agree

2.8 Variables Used in the Study

Independent Variables:

- Perceived Accuracy

  Includes:

  - Recommendations are relevant to user interests
  - AI recommendations correctly predict user preferences
  - Irrelevant recommendations are rarely received
  - Recommendations improve over time based on user behaviour

- Privacy Concerns

  Includes:

  - Concern about how personal data is used
  - Feeling that privacy is compromised due to AI tracking
  - Hesitation to interact due to privacy concerns
  - Concern about misuse of personal information
  - Discomfort in sharing data for recommendations

- Usefulness

  Includes:

  - Helps discover useful content
  - Saves time in searching
  - Improves overall social media experience
  - Helps in purchase decision making
  - Provides valuable and informative content

- Ease of Use

Includes:

- Easy interaction with recommended content
- Simple exploration of suggested content
- Easy understanding of AI recommendations
- Effortless navigation through recommended content
- Simple use of recommendation-related features

Dependent Variables:

- Customer Perception

  Includes:

  - Positive perception of AI recommendations
  - Belief that AI recommendations are beneficial
  - AI recommendations enhance social media experience
  - Overall favourable perception
  - AI recommendations are reliable

- Behavioural Intention

  Includes:

  - Likelihood of clicking recommended content
  - Engagement with recommended posts (like, share, comment)
  - Spending more time on recommended content
  - Purchase intention based on recommendations
  - Trusting recommendations enough to act on them

2.9 Tools Used for Analysis

2.9.1 Statistical Tools

The collected data was analysed using the following statistical tools:

- Descriptive Statistics

- Multiple Regression Analysis
- Simple Linear Regression
- ANOVA
- Independent Sample t-test

2.9.2 Software Tools

The following software tools were used for data analysis:

- R Programming Software
- Microsoft Excel
- Google Forms (for data collection)

R programming software was used for statistical analysis such as regression analysis, ANOVA, and t-test. Microsoft Excel was used for data cleaning, coding, and preliminary analysis. Google Forms was used to collect responses from respondents.

2.9.3 Analytical Tools Used Based on Objectives

Objective 1: Multiple Regression Analysis was used to examine the impact of perceived accuracy, privacy concerns, and usefulness on customer perception.
Objective 2: ANOVA and Independent Sample t-test were used to examine demographic differences in customer perception.
Objective 3: Simple Linear Regression was used to examine the influence of ease of use on customer perception.
Objective 4: Multiple Regression Analysis was used to analyze the impact of perceived factors on behavioural intention.

2.10 Limitations of the Study

- Limited sample size
- Time constraints
- Self-reported responses
- Study limited to Malappuram district

2.11 Chapter Summary

This chapter explained the research methodology adopted for the study. The study is based on primary data collected from respondents in Malappuram district. Stratified random sampling with proportionate allocation was used. Cochran's formula was applied to determine the sample size. Statistical tools such as regression, ANOVA, and t-test were used for analysis.

Chapter 3

THEORETICAL FRAMEWORK

3.1 Introduction

The theoretical framework forms the conceptual foundation of the research study. It helps in identifying the key variables and understanding the relationships between them. A well-defined theoretical framework provides direction for the study and supports the research objectives with relevant theories and concepts.

Artificial Intelligence (AI) based personalized recommendation systems have become an essential component of modern digital platforms. These systems are widely used in e-commerce websites, social media platforms, and online streaming services to enhance user experience by providing personalized suggestions. AI-based recommendation systems analyse user preferences, browsing patterns, purchase history, and behavioural patterns to generate relevant recommendations.

With the increasing use of AI-based recommendations, understanding customer perception and behaviour towards these systems has become important. While AI recommendations offer convenience and improved experience, users may also have concerns related to privacy, accuracy, and usefulness. Additionally, ease of use plays an important role in shaping customer perception.

This study focuses on examining the factors influencing customer perception and behaviour towards AI-based personalized recommendations. The study considers accuracy, privacy, usefulness, and ease of use as key independent variables. Customer perception and behaviour are considered dependent variables.

The theoretical framework of this study is based on the Technology Acceptance Model (TAM), which explains how users adopt and accept new technologies. TAM suggests that perceived usefulness and perceived ease of use influence user perception and behavioural intention. This model is widely used in studies related to technology adoption and user behaviour.

This chapter discusses AI-based personalized recommendations, Technology Acceptance Model, variables used in the study, proposed research model, and hypotheses development.

3.2 Artificial Intelligence-Based Personalized Recommendations

Artificial Intelligence-based personalized recommendation systems use machine learning algorithms and data analytics techniques to provide customized recommendations to users. These recommendations help users discover products, services, and content that match their preferences.

AI-based recommendation systems collect and analyse various types of user data, including:

- Browsing history
- Purchase behaviour
- Search patterns
- User preferences
- Demographic characteristics
- Interaction behaviour

Based on this data, AI algorithms generate recommendations that are relevant and useful for users. These recommendations are commonly seen in platforms such as online shopping websites, streaming services, and social media applications.

AI-based personalized recommendations provide several advantages:

- Improves user experience
- Saves time and effort
- Helps discover relevant content
- Enhances customer satisfaction
- Improves decision making
- Increases user engagement

Despite these advantages, users may also have concerns regarding AI-based recommendations. These concerns include:

- Privacy issues
- Data security concerns
- Accuracy of recommendations
- Over-dependence on technology

These factors influence customer perception and behaviour towards AI-based personalized recommendations. Therefore, it is important to analyse how these factors affect user perception and behaviour.

3.3 Technology Acceptance Model (TAM)

Technology Acceptance Model (TAM) was developed by Fred Davis in 1989. TAM is one of the most widely used models for studying user acceptance of technology.

According to TAM, users accept and adopt technology based on two main factors:

- Perceived Usefulness
- Perceived Ease of Use

These two factors influence user perception, which in turn affects behavioural intention and actual usage of technology.

The Technology Acceptance Model suggests the following relationship:

Perceived Ease of Use
↓
Perceived Usefulness
↓
User Perception
↓
Behavioural Intention
↓
Actual Usage

This model is relevant to AI-based personalized recommendations because users adopt AI recommendations based on their usefulness and ease of use.

3.4 Construct Variables of the Study

The present study uses six major construct variables namely Accuracy, Privacy, Usefulness, Ease of Use, Customer Behaviour, and Customer Perception. These constructs are measured using questionnaire statements with response options ranging from Strongly Disagree to Strongly Agree.
Each construct variable is explained below.

3.4.1 Accuracy

Accuracy refers to the extent to which AI-based personalized recommendations correctly match user preferences and interests. Accurate recommendations improve user trust and

perception toward AI-based systems.
In this study, accuracy is measured using the following statements:

- Recommendations are relevant to user interests
- AI recommendations correctly predict user preferences
- Irrelevant recommendations are rarely received
- Recommendations improve over time based on user behaviour

Higher accuracy of recommendations leads to improved customer perception and acceptance of AI-based personalized recommendations.

3.4.2 Privacy

Privacy refers to user concerns regarding personal data collection, tracking, and usage for AI-based recommendations. Privacy plays an important role in influencing customer perception.
In this study, privacy includes:

- Concern about how personal data is used
- Feeling that privacy is compromised due to AI tracking
- Hesitation to interact due to privacy concerns
- Concern about misuse of personal information
- Discomfort in sharing data for recommendations

Higher privacy concerns may negatively influence customer perception toward AI-based personalized recommendations.

3.4.3 Usefulness

Usefulness refers to the extent to which AI-based recommendations help users improve their experience and decision making.
In this study, perceived usefulness includes:

- Helps discover useful content
- Saves time in searching
- Improves overall social media experience

- Helps in purchase decision making
- Provides valuable and informative content

When users perceive recommendations as useful, their perception toward AI-based personalization becomes more positive.

3.4.4 Ease of Use

Ease of use refers to how easily users interact with AI-based personalized recommendation systems.
In this study, ease of use includes:

- Easy interaction with recommended content
- Simple exploration of suggested content
- Easy understanding of AI recommendations
- Effortless navigation through recommended content
- Simple use of recommendation-related features

Higher ease of use improves customer perception and acceptance of AI-based personalized recommendations.

3.4.5 Behavioural Intention

Behaviour Intention refers to user actions and engagement with AI-based personalized recommendations.
In this study, customer behaviour includes:

- Likelihood of clicking recommended content
- Engagement with recommended posts (like, share, comment)
- Spending more time on recommended content
- Purchase intention based on recommendations
- Trusting recommendations enough to act on them

Positive customer behaviour indicates acceptance and effectiveness of AI-based personalized recommendations.

3.4.6 Customer Perception

Customer perception refers to users' overall opinion and attitude toward AI-based personalized recommendations.
In this study, customer perception includes:

- Positive perception of AI recommendations
- Belief that AI recommendations are beneficial
- AI recommendations enhance social media experience
- Overall favourable perception
- AI recommendations are reliable

Positive perception leads to increased behavioural intention and engagement with AI-based personalized recommendations.

3.5 Proposed Research Model

The proposed research model is developed based on Technology Acceptance Model and research objectives.

- Accuracy influences perception
- Privacy influences perception
- Usefulness influences perception
- Ease of use influences perception
- Perception influences behaviour

Thus, customer perception acts as a mediating variable.

3.6 Hypotheses Development

Objective 1

- H1: Accuracy significantly influences customer perception
- H2: Privacy significantly influences customer perception
- H3: Usefulness significantly influences customer perception

Objective 2

- H4: Demographic factors significantly influence customer perception

Objective 3

- H9: Ease of use significantly influences customer perception

Objective 4

- H5: Accuracy significantly influences customer behaviour
- H6: Privacy significantly influences customer behaviour
- H7: Usefulness significantly influences customer behaviour
- H8: Perception significantly influences customer behaviour

3.7 Conceptual Framework

Accuracy, Privacy, Usefulness, Ease of Use

$\downarrow$

Customer Perception

$\downarrow$

Customer Behaviour

3.8 Chapter Summary

This chapter discussed the theoretical framework of the study. It explained AI-based recommendation systems, Technology Acceptance Model, variables, hypotheses, and conceptual framework. This provides a foundation for further analysis.

Chapter 4

DATA ANALYSIS AND INTERPRETATION

4.1 Introduction

This chapter presents the analysis and interpretation of the data collected for the study on customer perception of AI-based personalized recommendations. The data was collected through a structured questionnaire and analyzed using statistical techniques such as descriptive statistics, correlation analysis, regression analysis, ANOVA, and t-test. The purpose of this chapter is to examine the relationships between perceived accuracy, privacy, usefulness, ease of use, customer perception, and behavioural intention. The analysis is carried out based on the objectives of the study. A total of 250 responses were collected and analyzed using statistical software. The results obtained are presented systematically under each objective.

4.2 Data Collection Methodology

4.2.1 Survey Instrument Design

The data for this study was collected using a structured questionnaire designed to measure customer perception of AI-based personalized recommendations. The questionnaire consisted of multiple sections covering demographic variables and constructs related to AI-based recommendation systems.

The questionnaire included the following sections:

- Demographic Variables: Gender, Age Group, Educational Qualification
- Perceived Accuracy
- Privacy Concerns
- Perceived Usefulness
- Ease of Use
- User Perception
- Behavioural Intention

The responses were measured using a five-point response scale with the following options:

- Strongly Disagree
- Disagree

- Neutral
- Agree
- Strongly Agree

This response format was used to capture respondents' opinions and attitudes toward AI-based personalized recommendations.

The questionnaire items were designed based on previous research studies and adapted to suit the objectives of the study. Multiple items were used to measure each construct, and composite variables were created by calculating the average of the respective items.

4.2.2 Population of the Study

This study focuses on Malappuram District. Therefore, the population of the study comprises residents of Malappuram district.
According to Census 2011, the total population of Malappuram district is:

Population (N) = 4,112,920

Since the study focuses on AI-based personalized recommendations, the target population includes social media users and digital platform users within the district.

4.2.3 Sampling Technique, Sample Size Determination & Proportionate Allocation

Sampling Technique

This study adopted Stratified Random Sampling with Proportionate Allocation.
Stratified sampling was chosen to ensure representation across different age groups. The population was divided into three age strata:

- 18–25 years
- 26–35 years
- 36 years and above

Proportionate allocation was applied to ensure that each age group was represented according to its proportion.

Sample Size Determination

The required sample size was calculated using Cochran's Formula.

Cochran's Formula

$$n_0 = \frac{Z^2 \cdot p \cdot q}{e^2}$$

Where:

- n_0 = Sample size
- Z = Z-value at 95% confidence level (1.96)
- p = Estimated proportion of population (0.5)
- $q = 1 - p$ (0.5)
- e = Margin of error (0.05)

Substituting values:

$$n_0 = \frac{(1.96)^2 \cdot 0.5 \cdot 0.5}{(0.05)^2}$$

$$n_0 = \frac{3.8416 \cdot 0.25}{0.0025}$$

$$n_0 = \frac{0.9604}{0.0025}$$

$$n_0 = 384.16$$

Thus, the ideal sample size is approximately: $n_0 \approx 384$

Adjusted Sample Size

Due to time and resource constraints, the sample size was reduced to: n = 250
This corresponds to an approximate margin of error of 6%.
Although smaller than the ideal sample size, 250 respondents still provide reliable results for statistical analysis.

Proportionate Allocation

The sample size of 250 respondents was proportionately distributed across age groups:

Age Group	Percentage	Sample Size
18–25	45%	113
26–35	30%	75
36+	25%	62
Total	**100%**	**250**

Table 4.1: Age Group Distribution with Sample Size

4.2.4 Data Collection & Data Preparation

Data was collected through an online questionnaire distributed among residents of Malappuram district. The responses were collected over a specified period.

A total of 250 valid responses were collected and used for analysis.
The collected data was prepared for analysis using the following steps:

- Data cleaning
- Coding responses
- Creating construct variables
- Checking missing values
- Preparing dataset for statistical analysis

The following construct variables were created:

- Accuracy
- Privacy
- Usefulness
- Ease of Use
- Perception
- Behaviour

These variables were used for regression and statistical analysis.

4.3 Demographic Profile of Respondents

This section presents the demographic characteristics of the respondents included in the study. The demographic variables considered include age group, gender, and educational qualification. Understanding the demographic profile helped in interpreting customer perception toward AI-based personalized recommendations.

4.3.1 Age Group Distribution

The respondents were categorized into three age groups: 18–25 years, 26–35 years, and 36 years and above.
The distribution of respondents across these age groups is presented below.

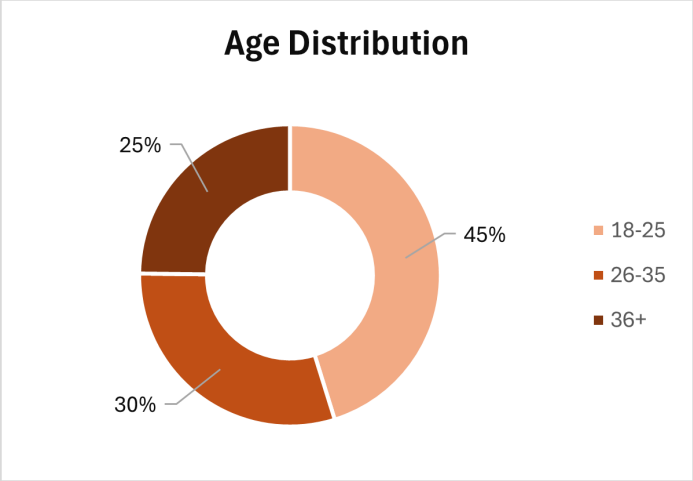

Figure 4.1: Age Distribution

Age Group	Number of Respondents	Percentage
18–25	113	45%
26–35	75	30%
36+	62	25%
Total	**250**	**100%**

Table 4.2: Age Distribution of Respondents

Interpretation

The majority of respondents belong to the 18–25 age group, accounting for 45% of the total respondents. This is followed by the 26–35 age group, which represents 30% of respondents, and the 36 years and above category, which accounts for 25%. This distribution

indicates that younger individuals form a larger proportion of the sample. Younger users are generally more active on digital platforms and are more exposed to AI-based personalized recommendations.

4.3.2 Gender Distribution

The gender distribution of respondents is presented below.

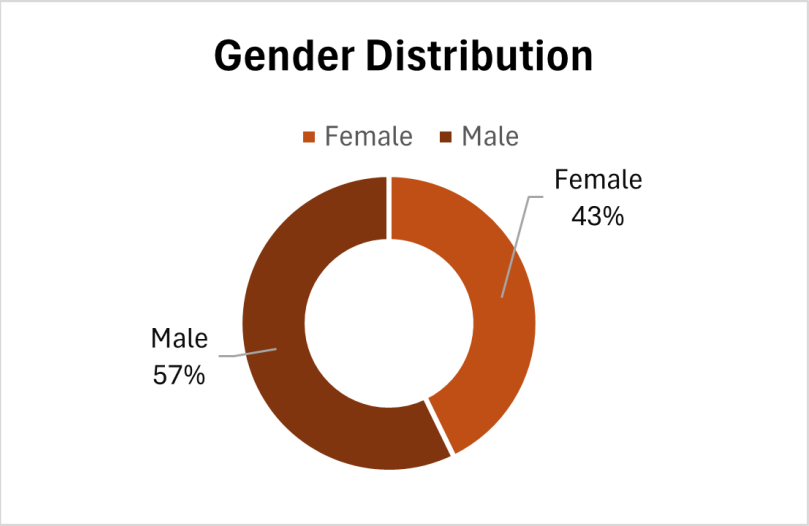

Figure 4.2: Gender Distribution

Gender	Number of Respondents	Percentage
Male	143	57.2%
Female	107	42.8%
Total	**250**	**100%**

Table 4.3: Gender Distribution of Respondents

Interpretation

The sample consists of 143 male respondents and 107 female respondents. Male respondents represent 57.2% of the total sample, while female respondents account for 42.8%.

This indicates that both genders are well represented in the study, allowing for meaningful comparison of customer perception across gender groups.

4.3.3 Education Distribution

The educational qualification of respondents is presented below.

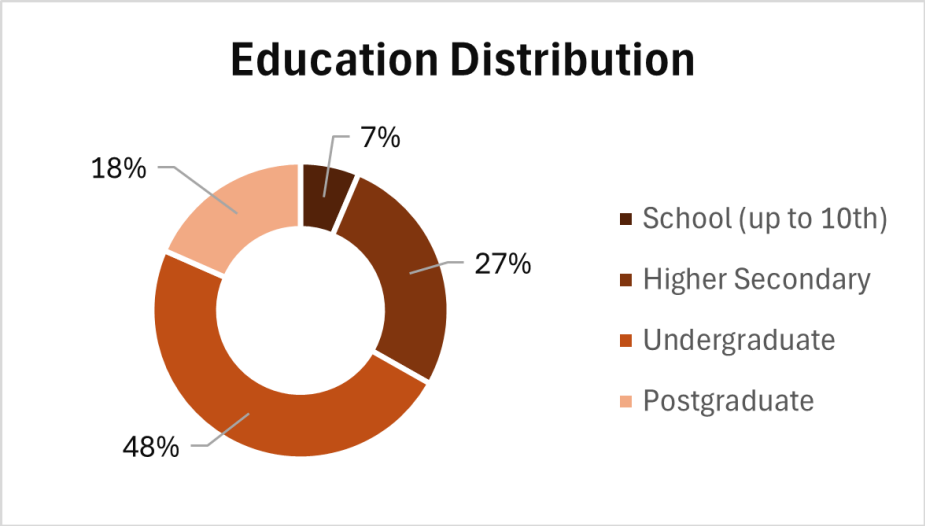

Figure 4.3: Gender Distribution

Educational Qualification	Number of Respondents	Percentage
Undergraduate	121	48.4%
Higher Secondary	67	26.8%
Postgraduate	46	18.4%
School (Up to 10th)	16	6.4%
Total	**250**	**100%**

Table 4.4: Educational Qualification of Respondents

Interpretation

The majority of respondents are undergraduate students, accounting for 48.4% of the total sample. Higher secondary respondents account for 26.8%, followed by postgraduate respondents at 18.4%. Respondents with school-level education account for 6.4%.

This distribution suggests that most respondents have moderate to high levels of education, which indicates familiarity with digital platforms and AI-based personalized recommendation systems.

4.4 Influence of Accuracy, Privacy and Usefulness on Customer Perception: Regression Analysis

4.4.1 Introduction

This objective aims to examine the influence of perceived accuracy, privacy concerns, and usefulness on customer perception of AI-based personalized recommendations. These factors are considered important in understanding how users evaluate and accept AI-driven recommendation systems.

Multiple regression analysis was conducted to determine how these independent variables influence customer perception.
Dependent Variable: Customer Perception
Independent Variables:

- Accuracy
- Privacy
- Usefulness

4.4.2 Hypothesis Formulation

H_{01}: Perceived accuracy does not significantly influence customer perception of personalized recommendations.
H_{11}: Perceived accuracy significantly influences customer perception of personalized recommendations.

H_{02}: Privacy concerns do not significantly influence customer perception of personalized recommendations.
H_{12}: Privacy concerns significantly influence customer perception of personalized recommendations.

H_{03}: Perceived usefulness does not significantly influence customer perception of personalized recommendations.
H_{13}: Perceived usefulness significantly influences customer perception of personalized recommendations.

4.4.3 Statistical Output

Multiple Regression Results

Variable	Estimate	Std. Error	t value	p value	Significance
Intercept	1.63229	0.36543	4.467	1.21e-05	Significant
Accuracy	0.31397	0.05465	5.745	2.71e-08	Significant
Privacy	-0.13186	0.06344	-2.078	0.0387	Significant
Usefulness	0.30515	0.06091	5.010	1.04e-06	Significant

Table 4.5: Regression Analysis Results

Model Summary

Statistic	Value
R-Squared	0.3951
Adjusted R-Squared	0.3877
F-Statistic	53.56
p-value	$< 2.2 \times 10^{-16}$

Table 4.6: Model Summary Statistics

4.4.4 Fitted Regression Model

General Model:

$$\text{Perception} = \beta_0 + \beta_1 \text{Accuracy} + \beta_2 \text{Privacy} + \beta_3 \text{Usefulness} + \epsilon$$

Estimated Model:

$$\text{Perception} = 1.632 + 0.314\,\text{Accuracy} - 0.132\,\text{Privacy} + 0.305\,\text{Usefulness}$$

4.4.5 Interpretation

The regression results indicate that perceived accuracy has a positive and significant influence on customer perception. This suggests that when AI recommendations are accurate, customers tend to develop a favorable perception.

Privacy concerns show a negative and significant relationship with customer perception.

This indicates that increased privacy concerns reduce customers' positive perception of AI-based personalized recommendations.

Perceived usefulness also shows a positive and significant influence on customer perception. This suggests that customers value recommendations that help them discover useful content and make better decisions.

The R-squared value of 0.3951 indicates that approximately 39.5% of the variation in customer perception is explained by accuracy, privacy, and usefulness.

The overall model is statistically significant, as indicated by the F-statistic and p-value.

4.4.6 Correlation Analysis HeatMap

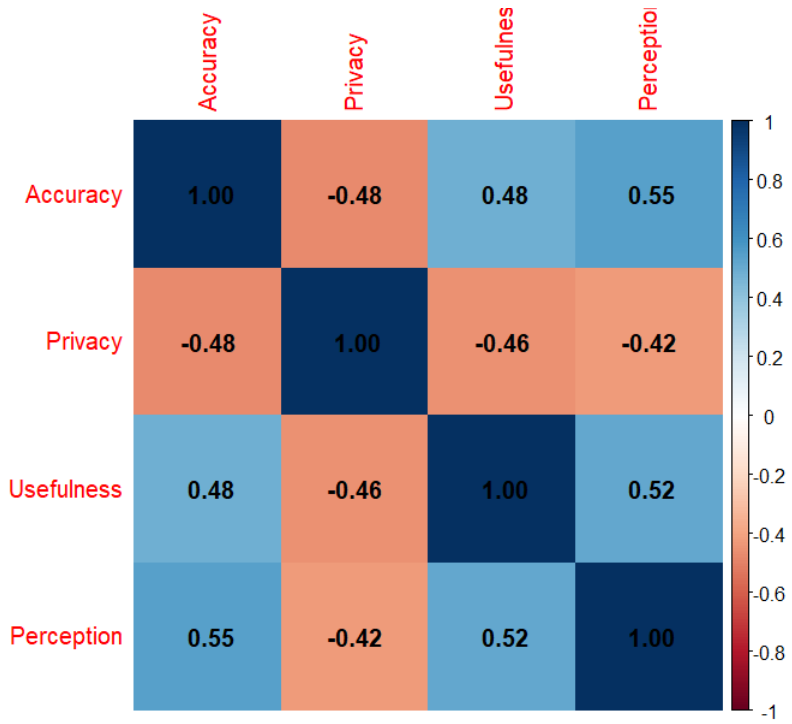

Figure 4.4: Correlation HeatMap

Interpretation

The correlation heatmap illustrates the relationship between the key variables of the study, namely Accuracy, Privacy, Usefulness, and Perception. A correlation heatmap is a graphical representation that uses colors to indicate the strength and direction of relationships between variables.

Color Variation in Heatmap

The heatmap uses a color gradient scale ranging from -1 to $+1$:

- Dark Blue $\rightarrow$ Strong Positive Correlation $(+1)$
- Light Blue $\rightarrow$ Moderate Positive Correlation
- White / Neutral $\rightarrow$ Weak or No Correlation (0)
- Light Orange / Red $\rightarrow$ Moderate Negative Correlation
- Dark Red $\rightarrow$ Strong Negative Correlation (-1)

Thus, color variation helps visually identify how strongly variables are related.

Interpretation of Each Relationship

Accuracy and Privacy (-0.48)

There is a moderate negative correlation between accuracy and privacy. This means when users perceive recommendations as more accurate, privacy concerns tend to decrease. The orange/red color indicates this negative relationship.

Accuracy and Usefulness (0.48)

Accuracy and usefulness show a moderate positive correlation. This indicates that when recommendations are accurate, users also perceive them as useful. The light blue color represents this positive relationship.

Accuracy and Perception (0.55)

Accuracy and perception have a moderate to strong positive correlation. This suggests that higher accuracy improves customer perception toward AI-based recommendations.

Privacy and Usefulness (-0.46)

Privacy and usefulness show a moderate negative correlation. This means when privacy concerns increase, users tend to perceive recommendations as less useful.

Privacy and Perception (-0.42)

Privacy has a negative correlation with perception. This indicates that higher privacy concerns reduce positive perception toward AI-based personalized recommendations.

Usefulness and Perception (0.52)

Usefulness and perception show a moderate positive correlation. This means that when users find recommendations useful, their perception becomes more positive.

4.4.7 Conclusion

The regression analysis indicates that perceived accuracy and usefulness significantly improve customer perception toward AI-based personalized recommendations. Privacy concerns negatively influence customer perception.
Therefore, the null hypotheses H_{01}, H_{02}, and H_{03} are rejected, and the alternative hypotheses are accepted. This indicates that perceived accuracy, privacy concerns, and usefulness significantly influence customer perception of AI-based personalized recommendations.

4.5 Influence of Demographic Factors in Customer Perception

4.5.1 Introduction

Customer perception toward AI-based personalized recommendations may differ across demographic groups. Factors such as age, gender, and educational qualification can influence users' familiarity with technology, trust in AI systems, and overall acceptance of personalized recommendations.

This objective examines whether customer perception varies across:

- Age Group (Generational Differences)
- Gender
- Educational Qualification

To analyze these differences, appropriate statistical tests were used:

- One-way ANOVA was used to examine differences across age groups.
- Independent sample t-test was used to compare perception between male and female respondents.
- One-way ANOVA was used to analyze differences across educational qualification

The dependent variable for this analysis is customer perception, while the independent variables include age group, gender, and educational qualification. This analysis helps identify whether demographic factors influence how customers perceive AI-based personalized recommendations.

4.5.2 Hypothesis Formulation

Age Group

H_{01}: There is no significant difference in customer perception across age groups.
H_{11}: There is a significant difference in customer perception across age groups.

Gender

H_{02}: There is no significant difference in customer perception between male and female respondents.
H_{12}: There is a significant difference in customer perception between male and female respondents.

Educational Qualification

H_{03}: There is no significant difference in customer perception across educational levels.
H_{13}: There is a significant difference in customer perception across educational levels.

4.5.3 Statistical Output

Age Group (ANOVA)

Source	Df	Sum Sq	Mean Sq	F value	p value
Age Group	2	0.07	0.0351	0.076	0.927
Residuals	247	113.97	0.4614		

Table 4.7: ANOVA Table for Age Group

Gender (t-test)

Gender	Mean Perception	t value	p value
Female	3.364		
Male	3.485	-1.392	0.165

Table 4.8: t-test Results for Mean Perception by Gender

Educational Qualification (ANOVA)

Source	Df	Sum Sq	Mean Sq	F value	p value
Education	3	0.7	0.2317	0.503	0.681
Residuals	246	113.3	0.4607		

Table 4.9: ANOVA Table for Education

4.5.4 Interpretation

The ANOVA results for age group show a p-value of 0.927, which is greater than 0.05. This indicates that there is no significant difference in customer perception across different age groups.

The gender t-test results show a p-value of 0.165, which is greater than 0.05. This indicates that there is no significant difference in customer perception between male and female respondents.

Similarly, the ANOVA results for educational qualification show a p-value of 0.681, which is greater than 0.05. This indicates that customer perception does not significantly differ across educational levels.

4.5.5 Visual Representation

Customer Perception by Age Group

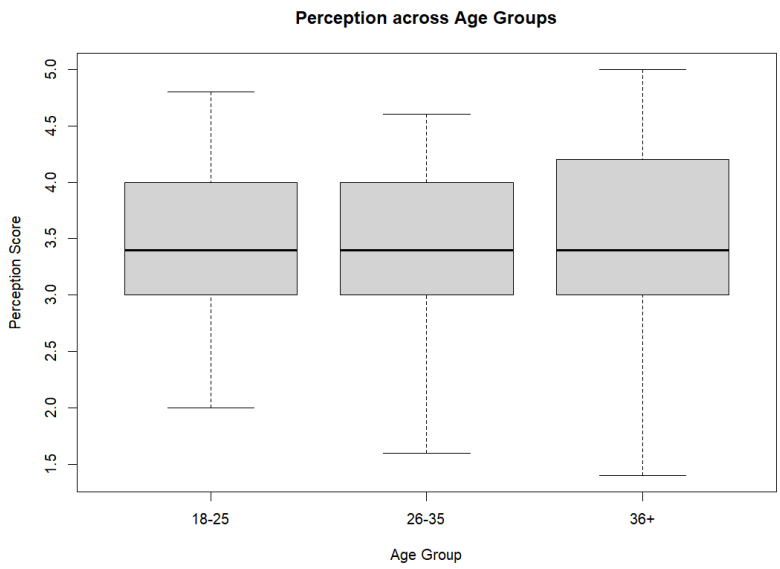

Figure 4.5: BoxPlot

Customer Perception by Gender

Figure 4.6: BoxPlot

Customer Perception by Education

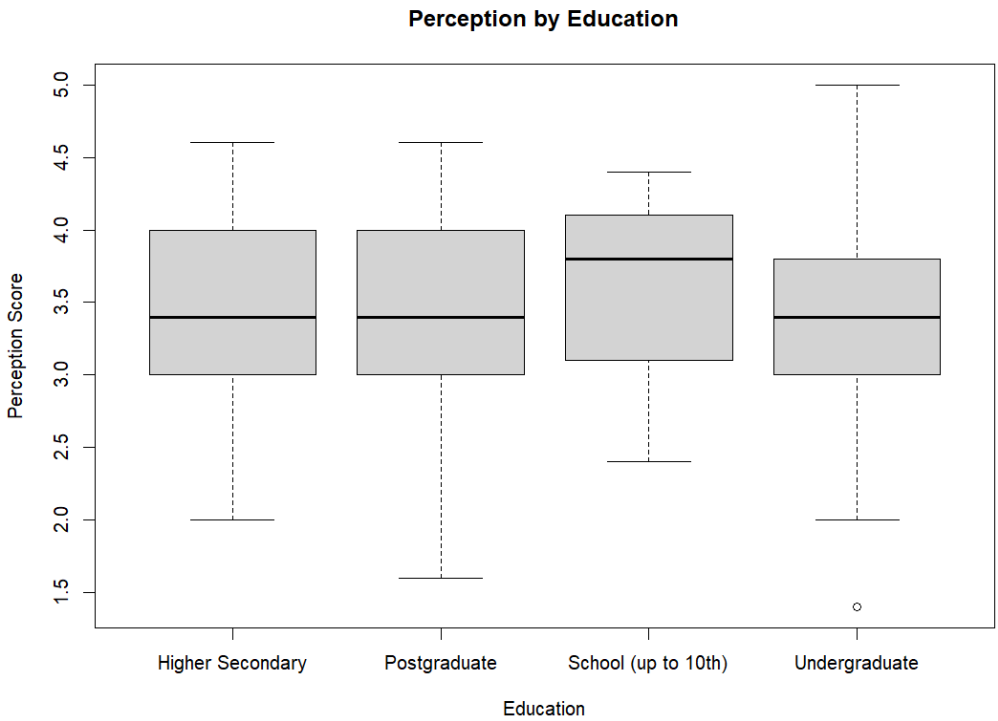

Figure 4.7: BoxPlot

Interpretation

The box plots show similar distribution of perception scores across age groups, gender, and educational levels. The median values and spread of perception scores appear nearly similar across groups, indicating minimal variation in perception among respondents.

4.5.6 Conclusion

The analysis indicates that there are no significant demographic differences in customer perception of AI-based personalized recommendations. Age group, gender, and educational qualification do not significantly influence customer perception.

Therefore, all null hypotheses are accepted. The findings suggest that customer perception toward AI personalization is consistent across different demographic groups.

4.6 Influence of Ease of Use on Customer Perception

4.6.1 Introduction

This objective aims to examine the influence of ease of use on customer perception of AI-based personalized recommendations. Ease of use is an important factor that determines how comfortably users interact with AI-driven systems. When recommendation systems are easy to understand, navigate, and interact with, users are more likely to develop positive perceptions.

Ease of use in this study was measured using the following indicators:

- Easy interaction with recommendations
- Easy exploration of recommended content
- Easily understandable recommendations
- Effortless navigation
- Simple and user-friendly features

A composite **Ease of Use** variable was constructed by taking the average of these indicators. Simple linear regression analysis was used to examine the influence of ease of use on customer perception.
Dependent Variable: Customer Perception
Independent Variable: Ease of Use

4.6.2 Hypothesis Formulation

H_0: Ease of use does not significantly influence customer perception.
H_1: Ease of use significantly influences customer perception.

4.6.3 General Regression Model

The general regression model is:

$$\text{Perception} = \beta_0 + \beta_1 \text{ Ease of Use} + \epsilon$$

Where:

- β_0 = Intercept
- β_1 = Regression coefficient
- ϵ = Error term

4.6.4 Summary Statistics for Ease of Use

Statistic	Value
Minimum	1.60
1st Quartile	3.00
Median	3.60
Mean	3.50
3rd Quartile	4.00
Maximum	5.00

Table 4.10: Descriptive Statistics

4.6.5 Statistical Output

Model Summary

Model	R	R Square	Adjusted R Square	Std Error
1	0.539	0.2905	0.2877	0.5712

Table 4.11: Model Summary

ANOVA Table

Source	Df	F value	p value
Regression	1	101.6	< 0.001
Residual	248		

Table 4.12: ANOVA Summary

Regression Coefficients

Variable	Coefficient	Std Error	t value	p value
Intercept	1.566	0.189	8.29	0.000
Ease of Use	0.533	0.053	10.08	0.000

Table 4.13: Regression Results

4.6.6 Fitted Regression Model

$$\text{Perception} = 1.566 + 0.533\,(\text{Ease of Use})$$

4.6.7 Interpretation

The regression results show that ease of use has a positive and significant influence on customer perception. The coefficient value of 0.533 indicates that an increase in ease of use leads to an increase in customer perception.

The R-square value of 0.2905 indicates that 29.05% of variation in customer perception is explained by ease of use. The model is statistically significant with an F-value of 101.6 and p-value less than 0.001.

This indicates that ease of use plays an important role in shaping customer perception toward AI-based personalized recommendations.

4.6.8 Visual Representation

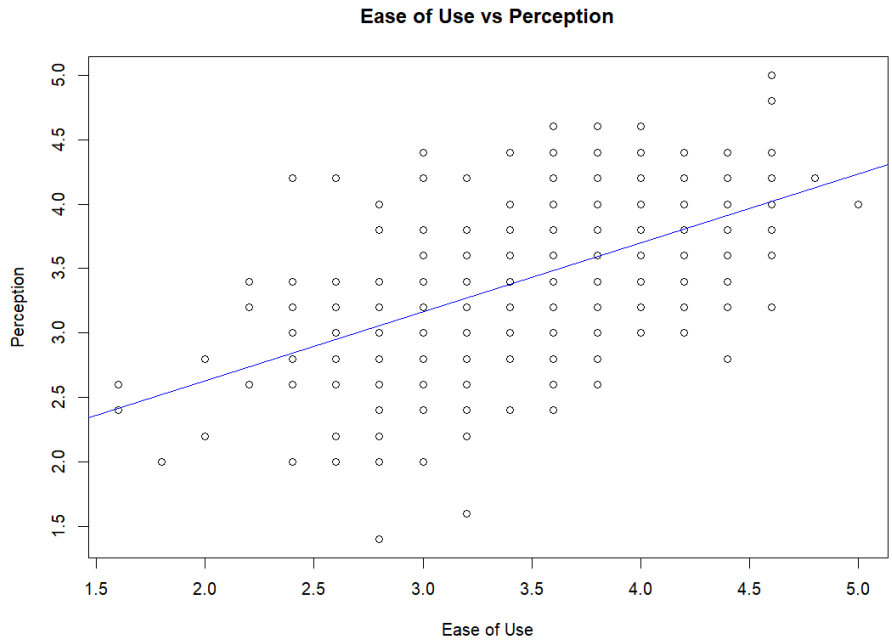

Figure 4.8: Scatter Plot

Interpretation

The scatter plot shows a clear positive relationship between ease of use and customer perception. As ease of use increases, customer perception also increases. Most of the data points are clustered along the upward-sloping regression line, indicating a strong positive association.

This suggests that when AI-based personalized recommendations are easier to interact with, understand, and navigate, customers tend to develop more favorable perceptions. The upward trend in the fitted regression line further supports the regression results, confirming that ease of use significantly influences customer perception.

4.6.9 Conclusion

The analysis indicates that ease of use significantly influences customer perception of AI-based personalized recommendations. Therefore, the null hypothesis is rejected and the alternative hypothesis is accepted.

4.7 Impact of Perceived Factors on Behavioural Intention

4.7.1 Introduction

This objective aims to analyse the impact of perceived factors on behavioural intention toward AI-based personalized recommendations. The study examines whether perceived accuracy, data privacy, usefulness, and overall perception influence customers' behavioural intention.

To examine this relationship, multiple linear regression analysis was used.

Dependent Variable: Behavioural Intention
Independent Variables:

- Privacy
- Usefulness
- Perception
- Accuracy

Multiple regression helps identify the combined effect of these factors on behavioural intention.

4.7.2 Hypothesis Formulation

H_0: Perceived factors do not significantly influence behavioural intention.
H_1: Perceived factors significantly influence behavioural intention.

4.7.3 Statistical Output

Regression Model Summary

Model	R	R Square	Adjusted R Square	Std. Error
1	0.708	0.5018	0.4937	0.5009

Table 4.14: Model Summary

ANOVA Table

Source	Df	F value	p value
Regression	4	61.7	< 0.001
Residual	245		

Table 4.15: ANOVA Summary for Regression Model

Regression Coefficients

Variable	Coefficient	Std Error	t value	p value
Intercept	1.538	0.359	4.281	0.000
Accuracy	0.205	0.055	3.731	0.000
Privacy	-0.203	0.061	-3.356	0.001
Usefulness	0.246	0.060	4.064	0.000
Perception	0.282	0.060	4.680	0.000

Table 4.16: Regression Coefficients

4.7.4 Fitted Regression Model

General Model:

$$\text{Behaviour} = \beta_0 + \beta_1(\text{Accuracy}) + \beta_2(\text{Privacy}) + \beta_3(\text{Usefulness}) + \beta_4(\text{Perception}) + \epsilon$$

Where:

- β_0 = Intercept
- $\beta_1, \beta_2, \beta_3, \beta_4$ = Regression coefficients
- ϵ = Error term

Variable Definitions:

- Behaviour = Behavioural Intention
- Accuracy = Perceived Accuracy
- Privacy = Privacy Concerns

- Usefulness = Perceived Usefulness
- Perception = Overall Customer Perception

Estimated Model:

$$\text{Behaviour} = 1.538 + 0.205(\text{Accuracy}) - 0.203(\text{Privacy}) + 0.246(\text{Usefulness}) + 0.282(\text{Perception})$$

4.7.5 Interpretation

The regression results indicate that perceived accuracy, usefulness, and perception have a positive and significant impact on behavioural intention. This means that as accuracy, usefulness, and perception increase, behavioural intention also increases.
Privacy shows a negative coefficient, indicating that higher privacy concerns reduce behavioural intention. This suggests that customers who are more concerned about data privacy are less likely to adopt AI-based personalized recommendations.
The R-square value of 0.5018 indicates that approximately 50.18% of variation in behavioural intention is explained by the independent variables included in the model.
The overall regression model is statistically significant, as the F-statistic is 61.7 with p-value less than 0.001.

4.7.6 Visual Representation

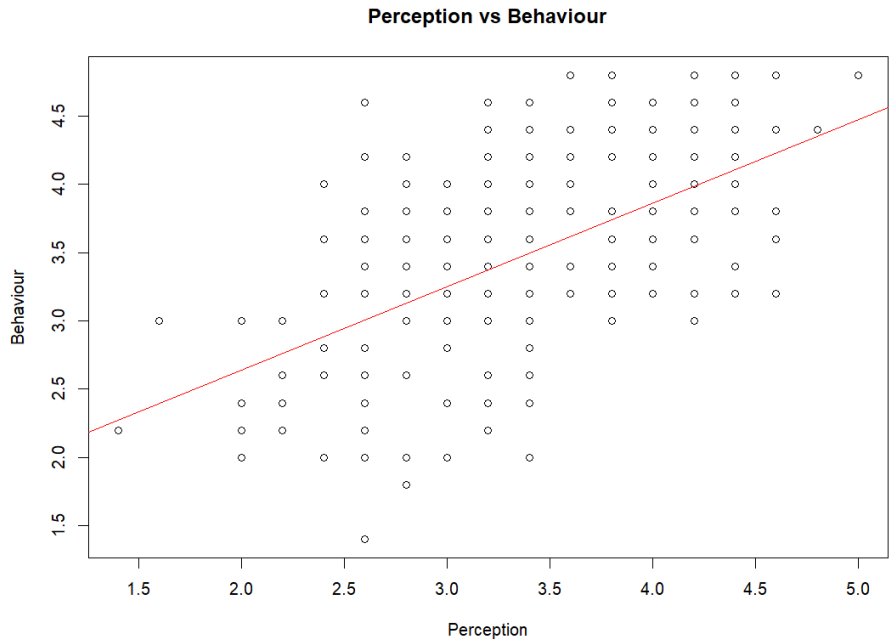

Figure 4.9: Scatter Plot

Interpretation

The scatter plot shows a positive relationship between customer perception and behavioural intention. As perception increases, behavioural intention also increases.
Most of the data points follow an upward trend along the regression line, indicating that customers with higher perception toward AI-based personalized recommendations are more likely to adopt and engage with such recommendations.
The fitted regression line further confirms that perception has a positive influence on behavioural intention. This supports the regression results obtained earlier.

4.7.7 Conclusion

The regression analysis indicates that perceived factors significantly influence behavioural intention. Accuracy, usefulness, and perception positively impact behavioural intention, while privacy negatively impacts behavioural intention.
Therefore, the null hypothesis is rejected. The findings indicate that perceived factors play an important role in influencing customer behavioural intention toward AI-based personalized recommendations.

4.8 Chapter Summary

This chapter presented the analysis and interpretation of data collected to examine customer perception and behavioural intention toward AI-based personalized recommendations. The analysis was conducted based on the four objectives of the study using appropriate statistical techniques such as multiple regression, ANOVA, and independent sample t-test.

Objective 1 examined the impact of perceived accuracy, privacy concerns, and usefulness on customer perception. Multiple regression analysis revealed that perceived accuracy and usefulness positively influenced customer perception, while privacy concerns showed a negative influence. The results indicated that customers develop favorable perceptions when recommendations are accurate and useful, whereas privacy concerns reduce positive perception.

Objective 2 explored generational and demographic differences in customer perception. The results of ANOVA and t-test showed no significant differences in perception across age group, gender, and educational qualification. This indicates that customer perception toward AI-based personalized recommendations is consistent across different demographic groups.

Objective 3 analyzed the impact of perceived factors on behavioural intention. Multiple regression analysis showed that accuracy, usefulness, and perception positively influence behavioural intention, while privacy concerns negatively affect behavioural intention. The results suggest that favorable perceptions and useful recommendations encourage customers to engage more with AI-based recommendations.

Objective 4 examined the influence of ease of use on customer perception. Simple linear regression analysis revealed that ease of use has a significant positive impact on customer perception. This indicates that easy interaction, simple navigation, and user-friendly features enhance customers' perception of AI-based personalized recommendations.

Overall, the findings of this chapter indicate that perceived usefulness, accuracy, ease of use, and overall perception play an important role in shaping customer behaviour toward AI-based personalized recommendations, while privacy concerns negatively influence customer acceptance. The results also suggest that demographic factors do not significantly affect customer perception, indicating widespread acceptance of AI-based personalized recommendations across different user groups

Chapter 5

FINDINGS, CONCLUSIONS AND SUGGESTIONS

5.1 Introduction

This chapter presents the major findings, conclusions, and suggestions based on the analysis conducted in the previous chapter. The findings are derived from statistical analysis performed to examine customer perception and behavioural intention toward AI-based personalized recommendations. The conclusions summarize the overall outcome of the study, and suggestions are provided to improve AI-based personalized recommendation systems.

5.2 Major Findings

Objective 1: Impact of perceived accuracy, privacy concerns, and usefulness on customer perception

- Perceived accuracy has a significant positive influence on customer perception.
- Perceived usefulness has a significant positive impact on customer perception.
- Privacy concerns show a negative influence on customer perception.
- Customers develop favorable perceptions when recommendations are accurate and useful.
- Higher privacy concerns reduce customer confidence.

Objective 2: Demographic differences in customer perception

- No significant difference across age groups.
- No significant difference between male and female respondents.
- Educational qualification does not significantly influence perception.
- Customer perception is consistent across demographic groups.
- AI-based recommendations are widely accepted across segments.

Objective 3: Influence of ease of use

- Ease of use has a significant positive impact on customer perception.
- Customers prefer systems that are easy to use.
- User-friendly design improves perception.
- Ease of use increases acceptance of AI systems.

Objective 4: Impact on behavioural intention

- Perceived accuracy positively influences behavioural intention.
- Perceived usefulness positively affects behavioural intention.
- Customer perception positively influences behavioural intention.
- Privacy concerns negatively influence behavioural intention.
- Customers engage more when recommendations are useful and accurate.

5.3 Overall Findings

- Accuracy and usefulness improve customer perception.
- Privacy concerns negatively influence perception and behavioural intention.
- Ease of use improves customer perception.
- Customer perception positively influences behavioural intention.
- Demographic factors do not significantly influence perception.
- Customers show positive acceptance toward AI-based recommendations.

5.4 Conclusion

The study examined customer perception and behavioural intention toward AI-based personalized recommendations. The results indicate that perceived accuracy, usefulness, and ease of use are key factors influencing customer perception, while privacy concerns negatively affect both perception and behavioural intention.
Customers prefer recommendations that are accurate, useful, and easy to use. When AI systems are perceived as reliable and beneficial, customers are more likely to engage with them. The study also finds that demographic factors do not significantly influence perception, indicating broad acceptance across user groups.
Overall, improving accuracy, usefulness, and ease of use while addressing privacy concerns can enhance adoption of AI-based personalized recommendations.

5.5 Suggestions

- Improve recommendation accuracy to enhance trust.

- Provide useful and relevant recommendations.
- Reduce privacy concerns through clear communication.
- Implement strong data protection measures.
- Design user-friendly interfaces.
- Provide transparency in AI recommendations.
- Offer personalization control options.
- Improve recommendation algorithms.
- Update AI models based on feedback.
- Increase awareness of AI benefits.

5.6 Scope for Future Research

- Include larger sample sizes.
- Conduct studies across different industries.
- Examine additional influencing factors.
- Conduct cross-country comparative studies.
- Perform longitudinal studies over time.

5.7 Limitations of the Study

- Limited sample size.
- Based on self-reported responses.
- Focuses on selected factors only.
- Findings may not be generalizable.

5.8 Chapter Summary

This chapter presented the findings, conclusions, and suggestions. The results indicate that perceived accuracy, usefulness, and ease of use positively influence customer perception, while privacy concerns negatively affect perception and behavioural intention. Suggestions and future research directions were also provided.

APPENDIX

QUESTIONNAIRE

1. Age:
   ○ 18–25
   ○ 26–35
   ○ 36 and above

2. Gender:
   ○ Male
   ○ Female
   ○ Other

3. Educational Qualification: ○ School Level (up to 10th)
   ○ Higher Secondary Level (up to 12th)
   ○ Undergraduate
   ○ Postgraduate
   ○ Other

   Please indicate your level of agreement with the following:

   Scale: ○ Strongly Disagree ○ Disagree ○ Neutral ○ Agree ○ Strongly Agree

4. The recommendations I receive on social media are relevant to my interests.
   ○ Strongly Disagree
   ○ Disagree
   ○ Neutral
   ○ Agree
   ○ Strongly Agree

5. AI recommendations correctly predict what I want to see.
   ○ Strongly Disagree
   ○ Disagree
   ○ Neutral
   ○ Agree
   ○ Strongly Agree

6. I rarely find irrelevant recommendations on my feed.
   ○ Strongly Disagree
   ○ Disagree
   ○ Neutral
   ○ Agree
   ○ Strongly Agree

7. The recommendations improve over time based on my behavior.
   ○ Strongly Disagree
   Strongly Disagree
   ○ Disagree
   ○ Neutral
   ○ Agree
   ○ Strongly Agree

8. I am concerned about how my personal data is used for recommendations.
   ○ Strongly Disagree
   ○ Disagree
   ○ Neutral
   ○ Agree
   ○ Strongly Agree

9. I feel my privacy is compromised when AI tracks my activity.
   ○ Strongly Disagree
   ○ Disagree
   ○ Neutral
   ○ Agree
   ○ Strongly Agree

10. I hesitate to interact with content due to privacy concerns.
    ○ Strongly Disagree
    ○ Disagree
    ○ Neutral
    ○ Agree
    ○ Strongly Agree

11. I worry about misuse of my personal information by social media platforms.
    ○ Strongly Disagree
    ○ Disagree
    ○ Neutral
    ○ Agree
    ○ Strongly Agree

12. I feel uncomfortable sharing data for personalized recommendations.
    ○ Strongly Disagree
    ○ Disagree
    ○ Neutral
    ○ Agree
    ○ Strongly Agree

13. AI recommendations help me discover useful content.
    ○ Strongly Disagree
    ○ Disagree
    ○ Neutral
    ○ Agree
    ○ Strongly Agree

14. The suggested content saves my time in searching.
    ○ Strongly Disagree
    ○ Disagree
    ○ Neutral
    ○ Agree
    ○ Strongly Agree

15. Recommendations improve my overall social media experience.
    ○ Strongly Disagree
    ○ Disagree
    ○ Neutral
    ○ Agree
    ○ Strongly Agree

16. I find the recommendations helpful for making purchase decisions.
    ○ Strongly Disagree
    ○ Disagree
    ○ Neutral
    ○ Agree
    ○ Strongly Agree

17. The content suggested is valuable and informative.
    ○ Strongly Disagree
    ○ Disagree
    ○ Neutral
    ○ Agree
    ○ Strongly Agree

18. It is easy to interact with recommended content.
    ○ Strongly Disagree
    ○ Disagree
    ○ Neutral
    ○ Agree
    ○ Strongly Agree

19. The platform makes it simple to explore suggested content.
    ○ Strongly Disagree
    ○ Disagree
    ○ Neutral
    ○ Agree
    ○ Strongly Agree

20. I find AI-based recommendations easy to understand.
    ○ Strongly Disagree
    ○ Disagree
    ○ Neutral
    ○ Agree
    ○ Strongly Agree

21. Navigating through recommended content is effortless.
    ○ Strongly Disagree
    ○ Disagree
    ○ Neutral
    ○ Agree
    ○ Strongly Agree

22. Using social media features related to recommendations is simple.
    ○ Strongly Disagree
    ○ Disagree
    ○ Neutral
    ○ Agree
    ○ Strongly Agree

23. I am likely to click on AI-recommended content.
    ○ Strongly Disagree
    ○ Disagree
    ○ Neutral
    ○ Agree
    ○ Strongly Agree

24. I often engage with recommended posts.
    ○ Strongly Disagree
    ○ Disagree
    ○ Neutral
    ○ Agree
    ○ Strongly Agree

25. I spend more time consuming recommended content.
    ○ Strongly Disagree
    ○ Disagree
    ○ Neutral
    ○ Agree
    ○ Strongly Agree

26. I am likely to purchase products suggested through recommendations.
    ○ Strongly Disagree
    ○ Disagree
    ○ Neutral
    ○ Agree
    ○ Strongly Agree

27. I trust recommendations enough to act on them.
    ○ Strongly Disagree
    ○ Disagree
    ○ Neutral
    ○ Agree
    ○ Strongly Agree

28. I have a positive perception of AI-based recommendations on social media.
    ○ Strongly Disagree
    ○ Disagree
    ○ Neutral
    ○ Agree
    ○ Strongly Agree

29. AI recommendations are beneficial to me.
    ○ Strongly Disagree
    ○ Disagree
    ○ Neutral
    ○ Agree
    ○ Strongly Agree

30. I believe AI recommendations enhance my social media experience.
    ○ Strongly Disagree
    ○ Disagree
    ○ Neutral
    ○ Agree
    ○ Strongly Agree

31. My overall perception of AI recommendations is favorable.
    - ○ Strongly Disagree
    - ○ Disagree
    - ○ Neutral
    - ○ Agree
    - ○ Strongly Agree

32. I feel AI recommendations are reliable.
    - ○ Strongly Disagree
    - ○ Disagree
    - ○ Neutral
    - ○ Agree
    - ○ Strongly Agree